

Bacterial ImuABC Damage-Induced Mutagenesis: A Review-Style Synthesis

A commissioned review of the SOS-regulated ImuA–ImuB–DnaE2 error-prone DNA-synthesis module

1. Executive Summary

Bacterial **ImuABC damage-induced mutagenesis** is a discrete, self-contained, SOS-regulated module for error-prone (mutagenic) DNA synthesis that operates in a very large fraction of bacteria that **lack the classical *E. coli* UmuDC/Pol V system**. The module is built from three co-inherited proteins encoded in a compact, damage-inducible cassette: **ImuA** (an accessory protein with distant homology to RecA/RadA-family recombinases), **ImuB** (a *catalytically dead* Y-family polymerase that acts as a molecular scaffold rather than a synthetic enzyme), and **DnaE2** (a duplicated C-family DNA polymerase III α -subunit that is the actual mutagenic catalyst). Together these constitute what is now often called the "**mutasome**." The system's essential job is to permit **damage-tolerant, low-fidelity DNA synthesis across lesions** that would otherwise stall replication — at the cost of introducing point mutations that drive genome diversification and, clinically, the evolution of antibiotic resistance.

The module's induction gate is the deeply conserved **RecA→LexA circuit**: DNA damage generates single-stranded DNA, RecA forms an activated nucleoprotein filament (RecA), *and this stimulates autoproteolysis of the LexA-family repressor, derepressing the SOS regulon — including the imuABC cassette*. A key mechanistic distinction from *E. coli* is that although RecA is required to induce the cassette, the downstream translesion-synthesis (TLS) step performed by ImuABC does not require a RecA nucleoprotein filament as a cofactor, unlike UmuDC/Pol V, which is absolutely RecA-dependent for its catalytic activation. Within the assembled mutasome, ImuB serves as the physical hub: it binds the **β -sliding clamp (DnaN)** through a canonical clamp-binding motif, engages **DnaE2**, and interacts with **ImuA** via a **RecA-N-terminal (RecA-NT) homology motif**. Disrupting any of these contacts abolishes mutagenesis, establishing the obligatory architecture $\text{ImuA} \leftrightarrow \text{ImuB} \leftrightarrow (\beta\text{-clamp, DnaE2})$.

This review defines the boundaries of the system (explicitly excluding nucleotide excision repair, homologous recombination, and constitutive chromosome replication), lays out the best current step-by-step model, catalogs lineage-specific variation (e.g., the ImuY–DnaE2 variant of *Deinococcus*, role-reassignment to DinB polymerases in *Streptomyces*, ImuA's RecA-inhibitory role in *Myxococcus*), traces its likely evolutionary origin as a repurposed replicative-polymerase paralog, and closes with the principal open questions — foremost the biochemistry of ImuA and the atomic architecture of the fully assembled mutasome. Throughout, we are explicit about which claims rest on strong genetic/biochemical evidence and which remain inferential or organism-specific.

2. Definition and Biological Boundaries

2.1 What the system is

The ImuABC module is a **reusable SOS-controlled cassette for mutagenic translesion DNA synthesis**. Its minimal functional unit comprises three gene products:

Component	Family / Nature	Role in the module
ImuA (ImuA', ImuY)	RecA/RadA-like accessory protein (non-catalytic)	First accessory step; interacts with ImuB; in some lineages modulates RecA
ImuB	Y-family polymerase fold, catalytically inactive	Central scaffold; binds β -clamp, DnaE2, and ImuA
DnaE2 (ImuC)	C-family Pol III α -subunit paralog	Error-prone catalytic polymerase; performs mutagenic synthesis

The cassette is transcriptionally controlled by the **RecA/LexA SOS gate**, and the biochemical output is **damage-tolerant, mutagenic DNA synthesis at stalled or damaged replication intermediates**.

2.2 What lies *outside* the boundary

Several neighboring processes are frequently conflated with ImuABC mutagenesis but are mechanistically distinct and should be treated separately:

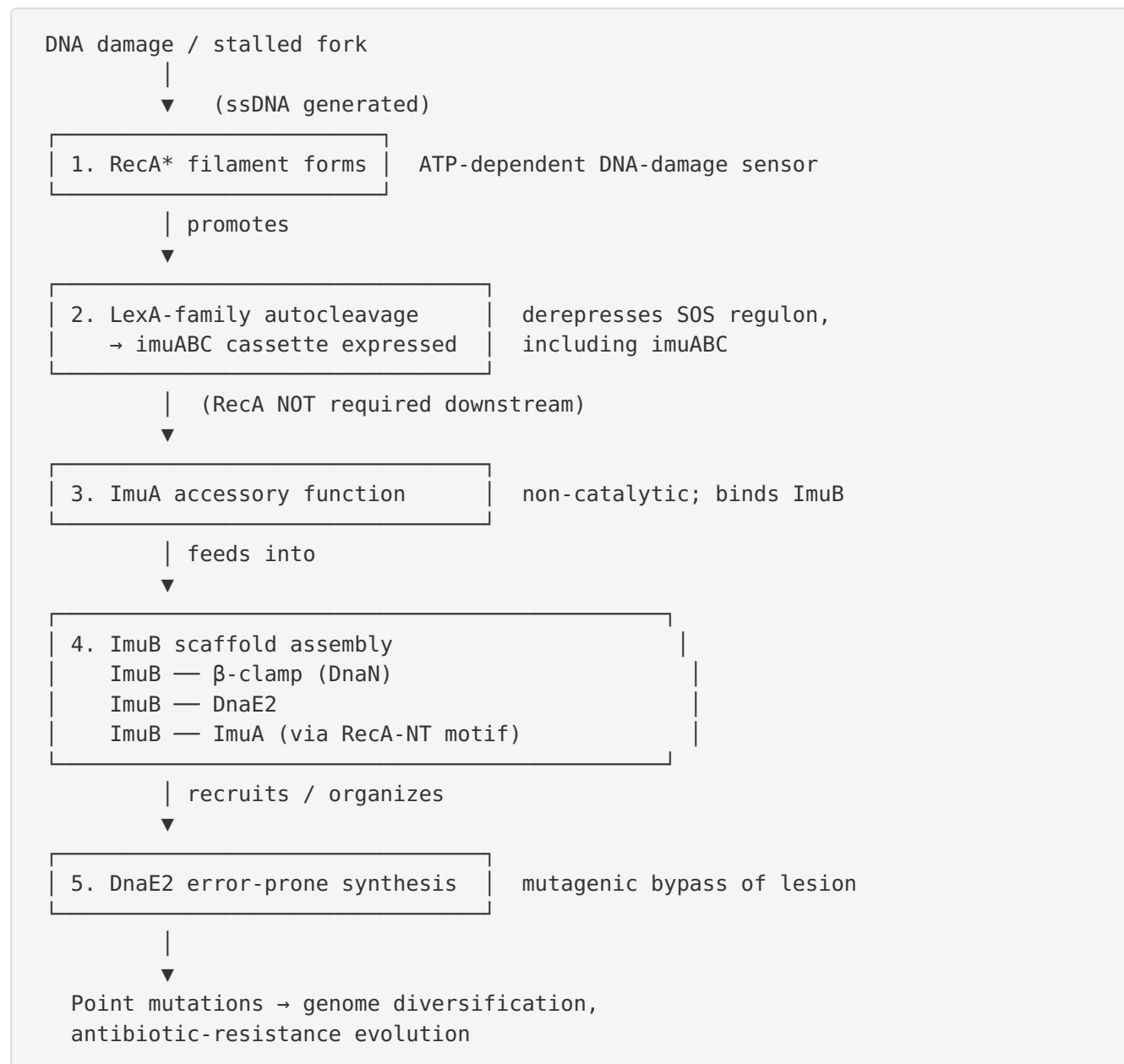
- **UmuDC / DNA polymerase V (Pol V):** The canonical *E. coli* SOS mutasome. It is a *different* solution to the same problem — a Y-family polymerase (UmuC) activated by cleaved UmuD' and an obligate RecA/ssDNA cofactor. *Many bacteria carry either umuDC or imuABC, and their distributions are largely complementary. ImuABC is the umuDC-independent* system.*
- **DinB / Pol IV (Y-family):** A separate, widely distributed TLS polymerase. In some organisms (notably *Streptomyces* and mycobacteria) DinB paralogs overlap with or substitute for DnaE2 in specific mutagenic outcomes, but DinB is not part of the ImuABC cassette.
- **Nucleotide excision repair (NER; UvrABC), base excision repair (BER), and homologous recombination (HR):** These are error-free or recombinational repair pathways. They can act upstream or in parallel (e.g., NER can generate or process substrates), but they are outside the module. One instructive boundary case in *Caulobacter* is **MmcB**, an endonuclease proposed to *create* substrates for ImuABC-mediated TLS patches — a feeder activity, not a module component.
- **Constitutive chromosomal replication by the primary replisome (DnaE1/PolC):** DnaE2 is a *second*, non-essential α -subunit copy dedicated to damage responses; the essential replicative polymerase is a separate protein.

2.3 Competing definitions and nomenclature

The literature is terminologically noisy. The same cassette appears under several names: **imuABC**, **imuAB-dnaE2**, and **imuC = dnaE2** (i.e., "ImuC" and "DnaE2" are the same protein in *Pseudomonas* and *Caulobacter* usage). *Mycobacterium* uses **imuA' (Rv3395c)** and **imuB (Rv3394c)**. *Deinococcus deserti* carries an **ImuY-DnaE2** variant. This synonymy is a genuine source of cross-organism confusion and must be tracked carefully when comparing studies.

3. Mechanistic Overview

3.1 Best current step-by-step model



3.2 Obligatory, conditional, and accessory steps

- **Obligatory:** (i) SOS induction via RecA→LexA is required to express the cassette at functional levels; (ii) all three proteins — ImuA, ImuB, DnaE2 — are individually required for induced mutagenesis; (iii) the ImuB–β-clamp interaction and the ImuB–ImuA (RecA-NT)

interaction are both essential. DnaE2's catalytic residues are essential (catalytic-dead DnaE2 phenocopies deletion).

- **Conditional/lineage-specific:** The precise role of ImuA (pure scaffold vs. RecA-modulator) appears to vary; a RecA-independent TLS step is demonstrated in *Caulobacter* but should not be assumed universal.
- **Accessory/feeder:** Lesion-processing endonucleases (e.g., MmcB for mitomycin-C damage), NER coordination in non-replicating cells, and nucleoid-associated proteins that promote polymerase exchange are supporting, not core, activities.

3.3 The critical mechanistic contrast with Pol V

The single most important mechanistic insight distinguishing ImuABC from the textbook *E. coli* system is that **ImuABC-mediated TLS does not require a RecA nucleofilament as a catalytic cofactor**. In *Caulobacter crescentus*, an operator-constitutive *imuABC* mutation rescues mutagenesis even in a *recA*-null background — proving that once the cassette is expressed at SOS-induced levels, ImuABC are the *only* genes needed for TLS, and RecA's role is confined to induction. This is mechanistically opposite to UmuDC/Pol V, whose catalytic activity is strictly dependent on a RecA*/ssDNA filament.

4. Major Molecular Players and Active Assemblies

4.1 The ImuA'-ImuB-DnaE2 mutasome is the umuDC-independent SOS mutagenesis module

Two foundational genetic systems anchor this claim. In *Caulobacter crescentus*, an operon of two hypothetical genes plus *dnaE2* (a second copy of the Pol III catalytic α -subunit) is **damage-inducible in a recA-dependent manner** and is **responsible for most UV- and mitomycin-C-induced mutations**; all three genes are required for error-prone lesion processing, and *umuDC* orthologs are absent (PMID: 15886391). In *Mycobacterium tuberculosis*, *imuA'* (Rv3395c) and *imuB* (Rv3394c) are **individually essential** for induced mutagenesis and damage tolerance, and DnaE2 catalytic-residue point mutants phenocopy the *dnaE2* deletion (PMID: 20615954). Together these establish that the cassette is a genuine three-component functional unit and the principal source of inducible mutagenesis in organisms lacking Pol V.

"an operon composed of two hypothetical genes and dnaE2, encoding a second copy of the catalytic subunit of Pol III, is damage inducible in a recA-dependent manner; and is responsible for most ultraviolet (UV) and mitomycin C-induced mutations in C. crescentus"
— PMID: 15886391

4.2 ImuB is a catalytically dead Y-family polymerase acting as a scaffold

ImuB retains the structural hallmarks of the Y-family polymerase fold but **lacks the conserved active-site residues required for catalysis** — it cannot itself synthesize DNA (PMID: 20615954). Instead it functions as the central organizing hub of the mutasome. Yeast two-hybrid and mutational studies show ImuB interacts with ImuA', with DnaE2, and with the **β -sliding clamp; disrupting the ImuB- β -clamp interaction significantly reduces induced mutagenesis and damage tolerance, phenocopying the deletion mutants** (PMID: 20615954). More recently, a **RecA-N-terminus (RecA-NT) homology motif within ImuB** was shown to be critical for the ImuB-ImuA' interaction (key residues L378/V383) (PMID: 39706264). The emerging picture is of ImuB as a "dead polymerase turned adaptor," using its clamp-binding motif to dock at the replication machinery and its RecA-NT motif to recruit ImuA, thereby positioning DnaE2 at the lesion.

"Despite retaining structural features characteristic of Y-family members, ImuB homologs lack conserved active-site amino acids required for polymerase activity" — PMID: 20615954

"RecA-NT is critical for the interaction of ImuB with ImuA" — PMID: 39706264

4.3 The induction gate is RecA-activated LexA autocleavage, but downstream TLS does not require a RecA filament

The upstream gate is the universal SOS switch: **RecA filaments stimulate the autocleavage of LexA**, the repressor of >50 SOS genes, activating the response (PMID: 36598938). The decisive downstream result comes from *Caulobacter*: an operator-constitutive *imuABC* mutation **rescues mutagenesis in a recA background**, showing that at SOS-induced expression levels ImuABC are the *only* genes required for TLS, and that **ImuABC-mediated TLS does not require RecA** — **unlike umuDC-dependent mutagenesis in E. coli** (PMID: 28938097). This cleanly separates RecA's regulatory role (induction) from any catalytic role (none, for ImuABC).

"the presence of the operator-constitutive mutation rescues mutagenesis in a recA background, indicating that imuABC are the only genes required at SOS-induced levels for translesion synthesis (TLS) in C. crescentus. Furthermore, these data also show that TLS mediated by ImuABC does not require RecA, unlike umuDC-dependent mutagenesis in E. coli"
— PMID: 28938097

4.4 DnaE2 is the catalytic mediator of inducible mutagenesis (founding evidence)

The founding study establishing DnaE2's role is Boshoff et al. (*Cell*, 2003). Using *M. tuberculosis*, they showed that UV irradiation increases mutation frequency in survivors; that *dnaE2* is upregulated by multiple DNA-damaging agents and during mouse infection; and that **loss of DnaE2 reduces post-UV survival and virulence in mice and reduces the emergence of drug resistance in vivo**. They concluded that **DnaE2 — a second replicative Pol III α copy, and *not* a Y-family polymerase — is the primary mediator of inducible mutagenesis** (PMID: 12705867). This paper is also the origin of the evolutionary framing: the "unresolved enigma" of multiple DnaE copies in pathogens and symbionts is resolved by one copy being repurposed for error-prone repair synthesis.

"Our data suggest that DnaE2, and not a member of the Y family of error-prone DNA polymerases, is the primary mediator of survival through inducible mutagenesis and can contribute directly to the emergence of drug resistance in vivo" — PMID: 12705867

"The presence of multiple copies of the major replicative DNA polymerase (DnaE) in some organisms, including important pathogens and symbionts, has remained an unresolved enigma. We postulated that one copy might participate in error-prone DNA repair synthesis"
— PMID: 12705867

4.5 Summary table of core players

Player	Family / motif	Activity	Essentiality (evidence)	Key PMIDs
RecA	RecA recombinase	ATP-dependent damage sensor; forms RecA*	Required for induction only	36598938, 28938097
LexA	LexA repressor (winged HTH)	Transcriptional repressor; autocleaves	Gate for cassette expression	36598938, 20703307
ImuA / ImuA' / ImuY	RecA/RadA-like accessory	Non-catalytic accessory; ImuB partner; RecA modulator (some lineages)	Essential (M. tb)	20615954, 34190612
ImuB	Y-family fold, dead	Scaffold; binds β -clamp, DnaE2, ImuA (RecA-NT)	Essential; clamp/ImuA contacts essential	20615954, 39706264
DnaE2 / ImuC	C-family Pol III α paralog	Error-prone catalytic TLS polymerase	Essential; catalytic residues required	12705867, 15886391
β -clamp (DnaN)	Sliding clamp	Processivity/docking platform for ImuB	Interaction essential	20615954

5. Evolutionary and Cell-Biological Variation

5.1 Clinical relevance and lineage variation

The cassette is **widespread but modular**, and its clinical importance is well documented. In mycobacteria, DnaE2 generates a **distinct mutational signature** and mediates the emergence of **rifampicin and fluoroquinolone resistance**, including in **antibiotic persisters** ([PMID:](#)

35918328, PMID: 35156855). DinB1 and DnaE2 have distinct but partially overlapping roles in substitution versus frameshift mutagenesis (PMID: 35918328). Across lineages the cassette varies in composition and naming:

- ***Deinococcus deserti*** uses an **ImuY–DnaE2** variant clustered under LexA control — "*imuY and dnaE2 form a gene cluster similar to a widespread RecA/LexA-controlled mutagenesis cassette*" (PMID: 19703105).
- ***Pseudomonas aeruginosa* / *P. putida** *ImuC (=DnaE2) contributes to ciprofloxacin resistance and alkylation-damage tolerance** (PMID: 28118378, PMID: 37625357).
- ***Myxococcus xanthus*** **ImuA inhibits RecA-mediated activity** to favor TLS, and DnaE2's functions require ImuA and ImuB — "*DnaE2 is an error-prone TLS polymerase, and its functions require ImuA and ImuB*" (PMID: 34190612).

"DinB1 promotes missense mutations conferring resistance to rifampicin, with a mutational signature distinct from that of DnaE2" — PMID: 35918328

"imuY and dnaE2 form a gene cluster similar to a widespread RecA/LexA-controlled mutagenesis cassette" — PMID: 19703105

5.2 Distribution across lineages

ImuABC is characteristic of bacteria that **lack** UmuDC/Pol V, and is prominent in **Actinobacteria** (mycobacteria, streptomycetes), **Alphaproteobacteria** (*Caulobacter*), **Deinococcus-Thermus**, and various **Beta/Gamma/Delta-proteobacteria** (*Pseudomonas*, *Myxococcus*). The distributions of *imuABC* and *umuDC* are broadly complementary, consistent with the idea that they are alternative, functionally interchangeable solutions to SOS mutagenesis. Analyses of DnaE paralog content tie the presence of a second *dnaE* (*dnaE2*) to genome GC-content variation and to broad ecological groupings, underscoring that DnaE2 is a major mutator whose evolutionary footprint is visible at the genome-composition level (PMID: 22230424).

5.3 Alternative routes to the same outcome

- **Polymerase substitution:** In *Streptomyces*, DnaE2 is **not** required for UV resistance/ mutagenesis; instead the two DinB paralogs (Pol IV) carry TLS/end-patching functions — a clear case of role-reassignment among specialized polymerases (PMID: 22006845).

- **Replication-independent action:** DnaE2 can act in **non-replicating cells**, coordinating with NER to enable damage survival — expanding the classical "at-the-fork" model of TLS (PMID: 33856342).
- **Polymerase-exchange regulation:** A nucleoid-associated protein promotes frequent exchange of the replicative polymerase, linking replisome dynamics to resistance emergence (PMID: 38591887).

5.4 Evolutionary origin

The deepest plausible origin of the catalytic core is a **gene duplication of the replicative Pol III α -subunit (dnaE)**, with one paralog (DnaE2) freed from essential replication duties and specialized for error-prone repair synthesis — precisely the hypothesis articulated in the founding *Cell* 2003 study (PMID: 12705867). ImuB likely derives from an ancestral Y-family polymerase that lost catalytic residues while retaining the fold and protein-interaction surfaces, converting an enzyme into an adaptor. ImuA's RecA/RadA-like fold points to recruitment from the ancient recombination machinery. The **most ancient and conserved elements** are therefore the RecA→LexA regulatory gate (shared with essentially all SOS systems) and the C-family polymerase fold of DnaE2; the **later elaborations** are the catalytic inactivation of ImuB and lineage-specific tuning of ImuA. For understanding the ancestral catalytic role, **DnaE2 in mycobacteria and *Caulobacter*** are the best-characterized representatives; for the regulatory gate, canonical *E. coli* LexA/RecA remain the reference despite the downstream mechanistic divergence.

6. Constraints, Dependencies, and Failure Modes

6.1 Obligatory ordering

1. **Damage → ssDNA → RecA* must precede LexA autocleavage.** No RecA activation, no cassette derepression.
2. **LexA autocleavage must precede cassette expression.** An intact LexA repressor keeps *imuABC* off (this is the "LexA cassette regulatory gate inhibits ImuA accessory step" relationship).
3. **ImuB scaffold assembly must precede productive DnaE2 synthesis.** ImuB must engage the β -clamp and ImuA for DnaE2 to be delivered to the lesion.

The operator-constitutive *Caulobacter* experiment ([PMID: 28938097](#)) demonstrates that step 1 can be *bypassed for the catalytic step* by artificially expressing the cassette — but under physiological conditions the ordering is strict because expression is gated.

6.2 Mutually exclusive / substrate-specific constraints

- **RecA's two roles are separable:** induction (required) vs. catalytic cofactor (not required for ImuABC). This rules out the otherwise-plausible "Pol V-like" model in which a RecA filament would be needed at the lesion for ImuABC catalysis.
- **Substrate specificity differs among polymerases:** In *Caulobacter*, **MmcB** is required specifically for mitomycin-C-induced mutagenesis but **not** for UV-induced mutagenesis, and it acts in the same pathway as ImuC/DnaE2 ([PMID: 26162909](#)). This indicates lesion-type-specific feeder activities upstream of a common TLS core.
- **Catalytic-dead DnaE2 is a null:** mutating DnaE2 active-site residues abolishes function, proving the catalytic requirement and ruling out a purely structural role for DnaE2 ([PMID: 20615954](#)).

6.3 Failure modes

- **Loss of any core component** (ImuA, ImuB, DnaE2) abolishes induced mutagenesis and reduces damage tolerance — and, in *M. tuberculosis*, reduces in vivo survival, virulence, and resistance emergence ([PMID: 12705867](#)).
- **Disrupting ImuB's β -clamp motif or its RecA-NT/ImuA interface** phenocopies deletion — the scaffold contacts are single points of failure ([PMID: 20615954](#), [PMID: 39706264](#)).
- **Blocking the SOS gate** (e.g., non-cleavable LexA, or nanobodies that trap LexA and prevent autoproteolysis) prevents cassette induction, a strategy being explored as an antibiotic adjuvant ([PMID: 36240773](#)).

7. Mechanistic Model — Consolidated Interpretation

The ImuABC module is best understood as a **regulatorily gated, structurally modular "mutasome" that trades fidelity for survival**. Its logic separates cleanly into a **regulatory layer** and an **effector layer**:

- **Regulatory layer (conserved, ancient):** DNA damage $\rightarrow$ ssDNA $\rightarrow$ RecA $\rightarrow$ LexA autocleavage $\rightarrow$ derepression of imuABC*. This is the same switch used by the entire SOS

regulon and is shared with UmuDC systems, prophage induction, and many stress responses.

- **Effector layer (modular, lineage-tuned):** ImuB, a dead Y-family polymerase, docks on the β -clamp at the stalled fork and, via its RecA-NT motif, recruits ImuA; the assembled scaffold then organizes DnaE2 to perform mutagenic synthesis across the lesion. Crucially, the effector step needs *no* RecA filament — the module is autonomous once expressed.

This architecture explains the system's evolutionary success: by co-opting a duplicated replicative polymerase (DnaE2) and converting an enzyme into an adaptor (ImuB), bacteria built a **portable, single-operon mutagenesis device** that plugs into the universal SOS gate. The cost is elevated mutation, which is precisely why the module is medically consequential: it is a **direct, druggable driver of antibiotic-resistance evolution**, from mycobacterial rifampicin/fluoroquinolone resistance (including in persisters) to *Pseudomonas* ciprofloxacin resistance.

The dependency chain, stated compactly, is:



8. Controversies and Open Questions

8.1 What does ImuA actually do?

ImuA is the least understood core component. Its RecA/RadA-like fold suggests a recombination-derived origin, but its biochemical activity is unresolved. In *Myxococcus* it **inhibits RecA-mediated activity** to favor TLS ([PMID: 34190612](#)), implying an active regulatory role; in mycobacteria it is framed primarily as an essential accessory/scaffold partner of ImuB via the RecA-NT motif ([PMID: 39706264](#)). Whether ImuA is fundamentally a RecA-antagonist, a loading factor, a nucleotide-dependent switch, or a pure adaptor — and whether this differs by lineage — is a central open question.

8.2 Atomic architecture of the assembled mutasome

There is currently **no high-resolution structure of the assembled ImuA–ImuB–DnaE2– β -clamp complex**. The stoichiometry, the geometry of DnaE2 handoff to the primer terminus, and how a catalytically dead Y-family fold positions a C-family polymerase are all inferred from interaction and mutational data rather than direct structural observation. Ongoing work on mutasome composition and recruitment ([PMID: 37530405](#), [PMID: 37034714](#)) is beginning to address this, but the structural picture remains a major gap.

8.3 How universal is RecA-independent catalysis?

The RecA-independence of the catalytic TLS step is firmly demonstrated in *Caulobacter* ([PMID: 28938097](#)). Whether this generalizes to all ImuABC-bearing lineages — especially those where ImuA modulates RecA (*Myxococcus*) — is not established, and cross-organism generalization should be made cautiously.

8.4 Division of labor among specialized polymerases

The overlapping and distinct roles of DnaE2 versus DinB paralogs (Pol IV) are organism-specific and incompletely mapped. In mycobacteria they have distinct mutational signatures ([PMID: 35918328](#), [PMID: 37141254](#)); in *Streptomyces*, DinB paralogs — not DnaE2 — carry the UV/TLS load ([PMID: 22006845](#)). A unified model of which polymerase handles which lesion, in which lineage, is lacking.

8.5 Evidence quality caveats

Much of the strongest mechanistic evidence comes from **two model systems** (*M. tuberculosis* and *C. crescentus*), plus focused studies in *Pseudomonas*, *Myxococcus*, and *Deinococcus*. Claims that mix data across these organisms should be read with care, because ImuA's role, RecA-dependence, and polymerase division of labor all appear to vary. The induction gate (RecA/LexA) is the best-conserved and most strongly supported feature; the catalytic role of DnaE2 and the scaffold role of ImuB are strongly supported genetically; the biochemistry of ImuA and the structural architecture are the weakest-supported and most inference-laden.

9. Limitations and Knowledge Gaps

- **No experimental data of our own:** This review is a literature synthesis; all conclusions rest on published genetic, biochemical, and interaction studies, weighted toward *M. tuberculosis* and *C. crescentus*.
 - **ImuA biochemistry is unresolved** — activity, cofactor requirements, and whether its RecA-modulatory role is general.
 - **No atomic structure** of the assembled mutasome; stoichiometry and DnaE2 handoff geometry are inferred.
 - **Generalization risk:** RecA-independence of catalysis, polymerase division of labor, and cassette composition all vary by lineage; single-organism results should not be extrapolated universally.
 - **Fidelity/signature data are sparse** outside mycobacteria; the mutational spectra of DnaE2 across diverse lineages are largely uncharacterized.
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10. Proposed Follow-up Experiments / Actions

1. **Cryo-EM / crystallography of the assembled ImuA–ImuB–DnaE2– β -clamp complex** on a primed lesion-containing substrate, to resolve stoichiometry and the polymerase handoff geometry.
2. **In vitro reconstitution of ImuABC TLS** with purified components to test, biochemically, the RecA-independence demonstrated genetically in *Caulobacter*, and to measure DnaE2 fidelity and lesion-bypass specificity directly.
3. **Systematic dissection of ImuA:** nucleotide-binding assays, RecA-filament competition assays, and structure determination to settle whether ImuA is an adaptor, a RecA-antagonist, or a switch — and whether this differs between *Myxococcus* and mycobacteria.
4. **Comparative mutational-signature profiling** of DnaE2 (and DinB paralogs) across multiple lineages (Actinobacteria, Alphaproteobacteria, Deinococcus, Pseudomonas) via whole-genome sequencing of damage survivors, to build a unified polymerase-vs-lesion map.
5. **Interface-targeted inhibitor screening:** given that disrupting the ImuB– β -clamp and ImuB–ImuA interfaces phenocopies deletion, screen for small molecules or peptides that

block these contacts as **anti-evolvability adjuvants** to suppress resistance emergence, complementing LexA-autocleavage inhibitors ([PMID: 36240773](#)).

6. **Test RecA-independence broadly:** replicate the operator-constitutive/*recA*-null rescue experiment in additional lineages to establish the generality (or limits) of the catalytic RecA-independence.
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11. Key References

PMID	Contribution
12705867	Founding evidence: DnaE2, not a Y-family polymerase, mediates inducible mutagenesis, in vivo survival, virulence, and resistance in <i>M. tuberculosis</i>
15886391	<i>Caulobacter</i> imuAB-dnaE2 operon: recA-dependent, damage-inducible, responsible for most UV/MMC mutations
20615954	ImuA'/ImuB essential accessory factors; ImuB is a catalytically dead Y-family scaffold; β -clamp interaction essential
28938097	Operator-constitutive rescue in <i>recA</i> null: ImuABC TLS does not require RecA, unlike Pol V
39706264	RecA-NT homology motif in ImuB mediates the essential ImuB–ImuA' interaction
37034714	RecA-NT motif essential for ImuA'–ImuB interaction and mutasome function
37530405	Composition and recruitment of the mycobacterial ImuA'–ImuB–DnaE2 mutasome
36598938	Structural basis of the RecA→LexA autocleavage SOS gate
20703307	LexA–DNA structure and SOS-box recognition
35918328	Distinct DnaE2 vs DinB1 mutational signatures; link to rifampicin resistance
37141254	Roles of mycobacterial DinB2 in frameshift/substitution mutagenesis
19703105	<i>Deinococcus</i> ImuY–DnaE2 variant of the RecA/LexA cassette
34190612	<i>Myxococcus</i> ImuA inhibits RecA to favor TLS; DnaE2 requires ImuA/ImuB
28118378	<i>Pseudomonas</i> ImuC/DnaE2 in alkylation-damage tolerance and mutagenesis
37625357	DinB, ImuBC, RpoS in <i>P. aeruginosa</i> ciprofloxacin-resistance mutations
35156855	DnaE2 mediates resistance evolution in mycobacterial persisters
22006845	<i>Streptomyces</i> : DinB paralogs, not DnaE2, carry TLS/end-patching
33856342	DnaE2 acts in non-replicating cells, coordinating with NER
26162909	MmcB feeds substrates to ImuC/DnaE2 specifically for MMC-induced mutagenesis
38591887	Nucleoid-associated protein promotes replicative-polymerase exchange and resistance
36240773	Nanobodies blocking LexA autocleavage as SOS-suppression/adjuvant strategy

PMID	Contribution
22230424	DnaE2 as a genome-scale mutator shaping GC-content variation

Prepared as a commissioned review synthesis. Evidence is weighted toward genetic and interaction studies in Mycobacterium tuberculosis and Caulobacter crescentus, with lineage-specific findings noted explicitly. The RecA/LexA induction gate and the catalytic role of DnaE2 are the best-supported features; the biochemistry of ImuA and the atomic architecture of the assembled mutasome remain the principal open questions.

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